Solar Lounge Project

Comprehensive Briefing Document

The "Solar Lounge" represents a high-premium architectural and engineering transformation of a residential rooftop in the North District of Israel, By integrating a 13.6 kWp solar array into a 96 square-meter luxury space.

Critical Takeaways:

- **Energy Balance :** The system is engineered to produce approximately 23,443 kWh annually, covering 141% of the home's 16,542kWh yearly consumption at worst case average.
- **System Configuration :** The high-efficiency hardware includes **20 Trina Vertex N 680W** bifacial panels (totaling **13.6 kWp**). The "brain" is a **10kW three-phase hybrid inverter** with an integrated ATS, supported by an optimized **14.3kWh LFP battery bank** (10.8kWh usable).
- **Technological Shift :** Traditional solar water boilers are replaced by a high-efficiency heat pump, powered by the solar array, to clear roof space and improve thermodynamic performance.
- **Financial Feasibility:** With a total estimated capital expenditure (CAPEX) of 95,000 ILS and annual savings/revenue of approximately 11,525 ILS, the project reaches its break-even point in roughly 8 years, offering minimum 17 subsequent years of **pure profit** over its 25-year lifespan.

Site and Solar Resource Analysis

The project is located in the North District of Israel at an elevation of 292 meters. Data from the Global Solar Atlas (GSA) indicates a high potential for photovoltaic (PV) development.

Solar Resource Data (Annual Averages) :

Parameter	Value
Direct Normal Irradiation (DNI)	2186.8 kWh/m²
Global Horizontal Irradiation (GHI)	2010.0 kWh/m²
Global Tilted Irradiation at Optimum Angle (GTI opta)	2227.4 kWh/m²
Optimum Tilt Angle	28° / 180° (South)
Specific PV Yield	1,695 kWh/kWp
Average Air Temperature	19.1 °C

Architectural Design

The design philosophy transforms 96 square meters (8m x 12m) of underutilized rooftop into a multi-functional "Solar Lounge".

- **Space Allocation :** The 20 PV panels occupy approximately 62.5 m^2, leaving nearly 34 m^2 of free space for architectural flexibility, This area features premium lounge seating, a professional teak billiard table, an olive wood dining table, and a marble bar.
- **The Pergola Structure :** The bifacial panels are integrated into a pergola. Almost 65% of the structure is covered by these panels which filter natural light while the other part remains open or can utilize skylights/aluminum louvers to maintain an airy feel.
- **The Angle Compromise :** the theoretical optimum tilt is 28°, the project chooses 26°.

System Engineering and Hardware Specifications

The project employs Tier-1 hardware to ensure longevity and maximum energy harvest.

Solar Array :

Trina Vertex N Series :

- Module Type : N-type i-TOPCon bifacial dual-glass monocrystalline.
- Unit Capacity : 680W (within a 675W–700W product range).
- Efficiency : Up to 22.5%.
- Durability : Resistant to harsh environments (salt, ammonia, sand), 1% degradation in the first year and 0.4% annually thereafter.
- Bifacial Advantage: Capable of up to 30% additional power gain from the backside depending on albedo.

Energy Management and Storage :

- **Hybrid Inverter (10kW Three-Phase) :** Acts as the system's brain, managing DC to AC conversion, battery charging, and grid interaction. It features an Automatic Transfer Switch (ATS) capable of switching to off-grid mode within 20 milliseconds during a power failure.
- **LFP Battery Bank (14.3 kWh) :** Utilizes Lithium Iron Phosphate technology, noted for being the safest in the market. It is rated for 6,000 to 8,000 charging cycles (15–20 years of daily use).
- **Dedicated Machinery Room :** A concrete niche was converted into a thermally insulated "technical bunker" with a steel door and active ventilation to protect the inverter and batteries from extreme heat.

Operational Logic and Electrical Infrastructure

The electrical architecture is designed for Absolute Independence, ensuring the home remains functional even during regional grid failures.

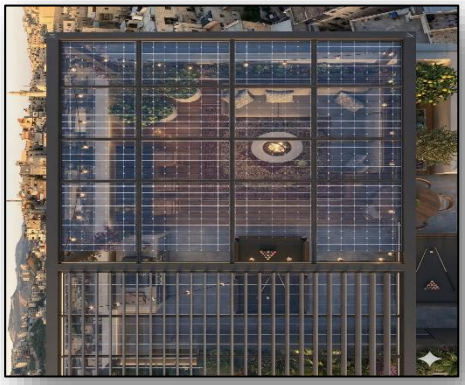
Load Management and Distribution :

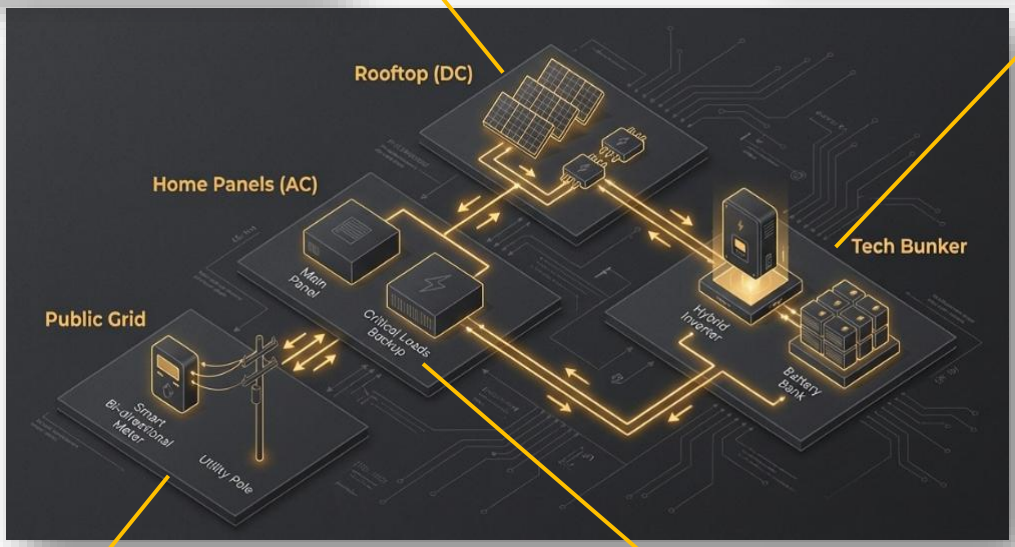
The system bifurcates the home's electricity into two distinct panels:

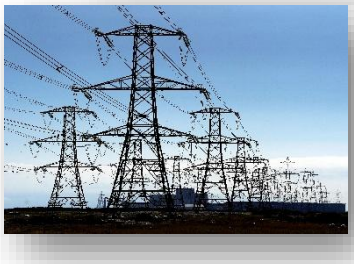
1. Main AC Panel : Supports non-critical, heavy loads such as ovens, central air conditioning, and washing machines. These disconnect during grid outages to preserve battery life.
2. Critical Loads Backup Panel : Connected to the inverter's backup output. This remains powered by the batteries and solar array during outages. Example for Connected Loads: Refrigerator, lights, internet routers, and the Heat Pump (ensuring continuous hot water).

Public Utility Interaction :

A Smart Bi-Directional Meter tracks energy imported from and exported to the grid.







Economic and Energy Financial Analysis

The project is structured as an investment with a high return on investment (ROI) and immediate property value appreciation.

Energy Math :

- Average Annual Consumption : 16,540 kWh.
- Annual Production : ~23,400 kWh.
- Surplus : ~7,000 kWh (used for reserve or future tools or sold to the grid).

Capital Expenditure (CAPEX) :

Category	Estimated Cost (ILS)
PV Panels & Hybrid Inverter	45,000
14.3kWh LFP Battery Bank	20,000
Heat Pump System	15,000
Infrastructure & Machinery Room Construction	15,000
Total Investment	**95,000**

Annual Savings and Revenue (OPEX) :

- Electricity Bill Offset: 16,542 kWh × 0.63 ILS/kWh = 10,421 ILS saved.
- Surplus Sale to Grid: 6,901 kWh × 0.16 ILS/kWh = 1,104 ILS revenue.
- Total Annual Benefit: 11,525 ILS.

Conclusion on ROI :

The project reaches its break-even point in approximately 8 years.

Given the 25-year lifespan of the panels, the owner can expect 17 years of **pure profit**, totaling roughly 195,000 ILS in net lifetime gains, alongside total energy security.